Leia Whitaker

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Education

Imperial College London **2024 – 2028**
MEng Design Engineering – First Class, Dean's List (Top 10%) *London, UK*

Tonbridge Grammar School **2022 – 2024**
International Baccalaureate – 45/45; HL Physics, Mathematics, Chemistry *Kent, UK*

Design Projects

CardClutch | *User Research, CAD, FEA, DFMA, Prototyping* **2026**

- Conducted user research (11 interviews, observation, journey mapping) across heritage institutions, iterating concepts to address slow, manual custom packaging in small collections where existing machinery is unaffordable.
- Led CAD development (team of 4) of a guided card cutter costing ~£115 per unit, using FEA to iterate geometry.
- Applied DFMA to cut part count from 44 to 30 (32% reduction) and optimise the casing for injection moulding.
- Built a working prototype that measures objects, generates box nets and guides cutting; exhibited at the Great Exhibition Road Festival, attracting investor interest.

Merry Mechanisms | *Mechanical Design, CAD, Electronics, C++, Pure Data* **2026**

- Designed and built a robot that frees Santa from a chimney when players solve a sensor-based puzzle, using CAD-designed mechanisms including a 20:1 gear train and linear actuator.
- Programmed state-based control logic coordinating 6 sensors, 2 actuators and audio to deliver interactive gameplay.
- Doubled vertical actuator force and refined mechanisms through 11 rapid prototypes.

Sustainable Aircraft Seat | *User Research, Sustainable Design, Materials* **2026**

- Translated passenger research into a modular comfort system retrofitting economy seat frames, integrating pressure zoning, lumbar support, active heating/cooling and tool-free disassembly.
- Used system mapping and benchmarking to design for full circularity, replacing PU foam with a mono-material flame-retardant PET stack and buy-back scheme; eco audit predicted ~45% lower embodied energy per kg.

TiltQuest | *User Research, CAD, Prototyping* **2025**

- Collaborated in a team to design a cooperative tilting-table game to address social integration barriers for newcomers, informed by interviews, co-design sessions and a 50-response survey; led the team's visuals, storyboards and renders.
- Developed and user-tested physical and digital subsystems across 14 prototypes; demonstrated language-independent gameplay in final co-design sessions, with players completing the game using visual cues alone.

Experience

Mechanics Team Member, Shell Eco-marathon – 1st in UK, 2nd Overall **2025 – 2026**
Imperial College London *London, UK*

- Redesigned shell fasteners to reduce weight by over 50% and improve alignment for aerodynamic performance; used FEA to optimise the strength-to-weight ratio of structural components, including the rear frame.

Front of House **July – Sept 2026**
Fortescue Inn *Salcombe, UK*

- Served up to 200 covers per shift in a front-of-house team, keeping service efficient and earning positive guest feedback.

Design Intern **June 2023**
Bond Bryan *Westerham, UK*

- Designed a user-centred yoga studio pavilion informed by site analysis, producing an Archicad 3D model and rendered walkthrough; presented to 15 staff and later selected by the practice as a showcase project.

Skills

- **CAD and Engineering:** Fusion 360, SolidWorks, Ansys (CFD, FEA), DFMA, 3D printing, laser cutting
- **Visual Communication and UI:** Adobe Creative Cloud, Figma, Blender
- **Programming:** MicroPython, C++, Python, JavaScript, HTML/CSS, MATLAB, Pure Data
- **Languages:** English (native), French (fluent)

Interests

Dance: Advanced-level vocational ballet; contemporary, acro and street dance; volunteer teacher of younger children.
Acting and Public Speaking: LAMDA Distinction.